

1. BUG: Classifier consistently returns "None" as intent

In EVERY test:

Query Flow Insights:

→  Classified as None → routed to crop_agent

That means:

- The classifier is being bypassed or
- The returned intent isn't being propagated correctly into SupervisorState

You never see:

```
intent = crop_agent
intent = general_rag
intent = weather_agent
```

Supervisor pre-classification is saving you here, but the classifier output is not being respected.

Impact

- Agents chosen incorrectly
- Wrong routing for mixed-intent queries
- Huge slowdown from unnecessary re-classifications

2. BUG: Supervisor always resets plan + executed_agents to []

Every single cycle of supervisor:

Current Plan: []

Executed Agents: []

Expected:

- On Turn 2 or 3 → executed_agents should persist
- plan should only be empty when new plan is actually done

Actual behavior:

- Supervisor forgets everything
- Even though session is loaded
- context → lost
- plan → lost
- executed_agents → lost

Impact

- Multi-agent plans DO NOT work
- Multi-turn reasoning breaks
- Clarifications cannot function
- Pronoun resolution cannot work ("there", "that crop")

This is one of your biggest bugs.



3. BUG: Location extraction happens... but never used anywhere

You can see:

LOCATION EXTRACTED: Bangalore → saved to user_context
But after that, nothing uses it.

The CropAgent immediately calls LLM to extract NPK/temperature/humidity → finds nothing → applies DEFAULTS.

So the system ignores location even though it was extracted correctly.

Impact

- Bangalore → outputs jute
- Pune → outputs jute
- Maharashtra → outputs jute
- Tamil Nadu → outputs jute
- Always jute

This is a *semantic pipeline bug*, not extraction.

4. BUG: CropAgent LLM extraction runs even when location-based query

Logs show:

```
crop_agent: Extracted parameters: {None...}  
crop_agent: Applied defaults...  
Calling crop recommendation tool  
Meaning:
```

- The agent **SHOULD** have switched to climate-based reasoning
- Instead it tries extraction on a query that clearly isn't numeric
- Defaults override actual context
- Model returns jute

Impact

Location-aware crop selection DOES NOT WORK.

5. BUG: LLM Clarification or Parameter Extraction Called Twice

CropAgent logs show two LLM calls back-to-back:

```
Node Execution: GenerateNode (807ms)  
Node Execution: GenerateNode (1271ms)  
This is duplicate LLM extraction.
```

Cause:

- CropAgent is calling `_extract_parameters_llm()` twice
- Or Supervisor is classifying + agent LLM extraction again
- Could also be caused by misrouting and re-processing

Impact

- Latency increases $\times 2$
- Costs double
- Still gets default values $\rightarrow$ still jute

6. BUG: Default parameters always overwrite extracted ones

The moment N/P/K/etc are missing:

```
Applied defaults for: ['N', 'P', 'K', 'temperature',  
'humidity', 'ph', 'rainfall']
```

This happens EVERY time climate data isn't filled in.

Even if location exists.

Even if LLM could infer temperature from known geography.

Defaults are ALWAYS applied last and override everything.

Impact

Massive consistent mispredictions.

7. BUG: Supervisor always sets next_agent = crop_agent for ANY query that contains a location

Log:

Location match found: Bangalore

Agent plan created: [], next_agent: crop_agent

But this rule is too naive:

You should first distinguish:

Query Type	Should Route To
“Weather in Bangalore?”	weather_agent
“Best crops in Bangalore?”	crop_agent

“Fertilizer for tomato in Bangalore?”	fertilizer_agent (+ maybe weather_agent first)
“Diseases common in Bangalore?”	general_rag (knowledge)

Supervisor's location logic is not conditioned on INTENT, only on extraction.

Impact

Misrouted agent calls.

8. BUG: conversation_history is not used in Supervisor

Your "contextual reference resolution" still doesn't work:

Test:

- Q1: Crops in Bangalore?
- Q2: What is the weather there?

Expected:

- "there" → resolve to Bangalore

Actual:

- Unhandled (not shown in logs, but already tested earlier)

Why?

- conversation_history is logged, but Supervisor ignores it
- No coreference logic
- No fallback to last known location

9. BUG: user_context is overwritten each turn instead of merged

Notice:

```
user_context = {}
location extracted → {'location': 'Bangalore'}
```

Next turn:

```
user_context resets → {}
```

So even if you store location once, the next turn wipes it out unless explicitly merged.

Impact

- Multi-turn state breaks
- Preference retention breaks
- Pronoun resolution impossible

10. BUG: CropAgent uses generic defaults that do not match India climate

Your defaults:

```
temp=25  
humidity=65  
rainfall=100  
These are not:
```

- Bangalore
- Pune
- Tamil Nadu
- Kerala
- Rajasthan

They're "generic tropical".

Impact

Jute becomes wrong default recommendation across the entire country.